

Vol 3 Chapter 12 — Bridge to Vol 4 and Vol 9

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2026-05-23 Saturday

Chapter 12 — Bridge to Vol 4 and Vol 9

This is the closing chapter of Vol 3. The volume has derived nuclear physics (magic numbers, shell model, SEMF) and atomic physics (orbital sequence, spectroscopy, atomic clocks, decay) from substrate primaries. This chapter cross-links to two adjacent volumes: Vol 4 (cosmology) and Vol 9 (condensed matter).

12.1 Cross-link to Vol 4 — BBN and cosmological nucleosynthesis

The nuclear-physics framework of Vol 3 anchors the cosmological derivations of Volume 4 Chapter 8 — Big Bang Nucleosynthesis. The substrate's nuclear scale and the substrate's BBN freeze-out temperature $T_c = N_{\max} \cdot 20/21 = 0.487$ MeV combine to produce the primordial light-element abundances at sub-percent precision (H, D, He) with the Li-7 problem resolved via the $\Delta g = 7$ substrate-genus correction.

12.2 Cross-link to Vol 9 — condensed matter and superconductivity

The atomic-spectroscopy framework of Vol 3 anchors the condensed-matter applications of Volume 9. Crystal lattice structure, electron band theory, magnetism, superconductivity — each inherits substrate K-type structure from the atomic orbital sequence we derived in Chapter 7. The substrate's $\kappa_{ls} = 6/5$ spin-orbit coupling and the $\alpha^{\text{BST primary}}$ exponent pattern carry through to condensed-matter calculations.

12.3 What comes next

Vol 4 begins the cosmological derivations (Newton's G and the cosmological constant already rewritten as recruiter chapters). Vol 5 develops quantum mechanics; Vol 6 thermodynamics; Vol 7 electromagnetism; Vol 8 classical mechanics; Vol 9 condensed matter; Vol 10 mathematical methods; Vol 11 generative geometry; Vol 12 chemistry; Vol 13 biology; Vol 14 information theory; Vol 15 methodology.

The substrate framework continues across every physics domain. The reader who has worked through Vol 3 has the apparatus for nuclear and atomic physics applications throughout the remaining volumes.

Where to look this up: BBN cross-link Vol 4 Ch 8. Condensed-matter cross-link Vol 9. For volume-wide synthesis: Volume 1 Chapter 11 observables reference.